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Growth and survivorship of American beech (*Fagus grandifolia* Ehrh.) seedlings in a northern hardwood forest following a mast event¹

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CLEAVITT, N. L., FAIRBAIRN, M., AND FAHEY, T. J. (Department of Natural Resources, Cornell University, Ithaca, NY 14853-3001). Growth and survivorship of American beech (*Fagus grandifolia* Ehrh.) seedlings in a northern hardwood forest following a mast event. J. Torrey Bot. Soc. 135: 328–345. 2008.—Despite the recognized importance of root sprout production in northern populations of American beech (*Fagus grandifolia* Ehrh.), further consideration of the relative role of regeneration from seed in this species appears warranted and timely. A careful examination of the literature revealed that seed- and sprouts-origin individuals co-occurred within all published study sites where the two were differentiated and that presence of a seedling bank may be the rule rather than the exception for American beech. We compared the effects of canopy gaps and root trenching on the growth and survival of a cohort of beech seedlings in Hubbard Brook Experimental Forest (HBEF) in northern New Hampshire, USA. Seedlings in gap plots became larger both above- and belowground. Foliar nitrogen concentrations were affected by both above- and belowground treatments and increased the most for seedlings in trenched gap plots. Seedling height and canopy treatment were important predictors of seedling survivorship with the highest survival in gap locations. The relationship between seedling height and survival was more complex. In the first two years, taller seedlings had a higher probability of dying, while after the second year taller seedlings were more likely to survive to be four-years old. The relatively high survivorship of seedlings at HBEF and their response to gap openings taken together with literature documentation of the long-term presence of seed-origin beech stems in the understory strongly suggest the presence of a beech seedling bank in this forest. This is one of a few manipulative studies on naturally regenerating American beech seedlings, and the first study to relate the likelihood of survival to seedling traits for this important species in the northern hardwood forest.

Key words: American beech, canopy gap, root trenching, seedling, survivorship.

American beech (*Fagus grandifolia* Ehrh.) has a multi-faceted regeneration strategy that includes seed masting and root suckering. The relative importance of seedling establishment

and vegetative root sprouting to maintenance of American beech populations in North America appears to vary geographically. Northern populations in areas that have been glaciated produce root sprouts (suckers) more frequently than southeastern piedmont and coastal plain populations (Kitamura and Kawano 2001). Indeed, recent work by Denk et al. (2005) has resurrected the idea that these northern populations represent the subspecies *grandifolia*, while the southern populations belong to subsp. *caroliniana*. Studies on the northern “gray” beech (subsp. *grandifolia*) indicate an increase in sprout prevalence on sites with shallow soils, sites that have been disturbed by mechanical damage (usually as a result of logging) and sites with more severe freeze-thaw cycles (Held 1983, Jones and Raynal 1988, Thurston et al. 1992). In a recent population genetics study of “gray” beech and “red” beech growing in close proximity in the Great Smoky Mountains, the prevalence of sprouting in both beech taxa was linked to site disturbance history rather than present day stem density within the sites (Morris et al. 2004). In general, root sprouting in American beech is a direct response to root injury.

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Regeneration by beech species in other parts of the world also includes the presence of a seedling bank. For instance, European beech (*Fagus sylvatica* L.) (Szwagrzyk et al. 2001) and Japanese beech (*F. crenata* Blume) (Nakashizuka 1987, Abe et al. 2005) both exhibit seedling banks. Marks and Gardescu (1998) have suggested that American beech should also be investigated for the presence of a seedling bank since this species possesses both shade tolerance and seed masting, which are common traits of other seedling bank species. Seed masting, the episodic increase in seed production, serves an important function in the maintenance of seedlings banks in sugar maple (Marks and Gardescu 1998), European beech (Szwagrzyk et al. 2001), and Japanese beech (Hiroki and Matsubasa 1995, Abe et al. 2005).

Regeneration of beech following the mortality of large adult beech to beech bark disease (BBD), a scale insect and fungus complex, has frequently been thought to involve mainly root sprouting. An influential paper by Houston (1975) documented an increased number of sapling sized beeches in 12 beech stands in Maine that had been devastated by beech bark disease (BBD) 30–40 years previous. Houston (1975) implied that these saplings were exclusively the result of increased regeneration by vegetative sprouting and many investigators have since followed his example (e.g., Runkle 1990, 2007, Fahey 1998, Forrester et al. 2003, Hane 2003). However, BBD has not always been followed by a dramatic increase in saplings (Twery and Patterson 1984, for sites in Massachusetts and New Hampshire). More importantly, a number of subsequent studies have shown that sprout-origin beech stem densities do not necessarily differ between stands as a result of BBD effects (Jones and Raynal 1986, 1987, Perrillo and Witter 2004). Indeed, Houston (1994) acknowledged that root sprout- and seed-origin individuals can both contribute to any increase in beech saplings recorded in response to BBD. Houston (2001) further reported that beech seedling densities were higher than sprout densities prior to harvesting treatments on a study site with a long history of BBD in north-central Maine.

Other aspects of beech population structure that call into question the dominance of sprout regeneration post-BBD include the persistence of high genetic diversity within populations

affected by BBD (Houston and Houston 2000) and maintenance of seed production per hectare in stands with some BBD resistant trees (McNulty and Masters 2004). Recent genetic studies examining within and between population diversity found that northern populations (subsp. *grandifolia*) contain as much genetic diversity as the southern populations (subsp. *caroliniana*) and concluded that sexual reproduction is a significant component of beech population genetics even in populations with more frequent root sprouting (Kitamura and Kawano 2001). Despite past emphasis on the importance of sprout production in regeneration by *Fagus grandifolia* supsp. *grandifolia*, further consideration of reproduction by seed appears warranted and timely. In particular, we seek to place information gained in this study in perspective with past studies of beech seedlings at Hubbard Brook Experimental Forest (HBEF) to provide a longer-term perspective on the possible role of beech seedlings in beech regeneration at this site.

Reviewing what is known regarding American beech seedling ecology is complicated by the use of “seedling” to denote a class size (often individuals shorter than 0.5 m) rather than to specifically refer to young beech individuals of seed-origin. For this reason we will use quotations as “seedling” when referring to literature that has used the term as a size-class. From the studies that have distinguished between sprout- and seed-origin stems, this synopsis of beech seedling ecology can be drawn: 1) seed- and sprout-origin stems co-occurred within all study sites (note especially Ward 1961, Held 1983, studies often cited for the importance of sprout regeneration), 2) multi-year persistence of seedlings were the rule not the exception (e.g., Forcier 1973, 1975, Thurston et al. 1992, Houle 1991, 1994, Houston 2001), 3) seed-origin stems were often more numerous than sprout-origin stems prior to tree harvest treatments (Thurston et al. 1992, Houston 2001), 4) seedlings were more numerous under overstory beech and in stands with higher beech basal area (BA) (Houston 2001, Messaoud and Houle 2006), 5) the youngest age classes of seedlings (0–2 yrs) suffered the highest mortality (Forcier 1973, 1975, Houle 1994), and 6) seedling growth was benefited by higher light levels, but not by the highest light levels with evidence for release of “seedlings” (Houston

2001, "seedling" studies: Poulson and Platt 1996, Beaudet and Messier 1998, Beaudet et al. 2000).

While studies on beech saplings suggest that this stage responds to increased light but not to root trenching or increased nutrients (Finzi and Canham 2000, Ricard et al. 2003), an earlier study on beech seedlings found that seedlings responded to nutrient additions at intermediate and higher light levels (Latham 1992). In plantation situations, European beech seedlings have been shown to benefit from root trenching treatments, which decreased competition for belowground resources from the overstory trees (Ammer 2002, German and Dutch studies cited in Löf et al. 2005). Nutrients were shown to affect growth but not survival of European beech seedlings, and nutrient effects were only apparent for individuals grown under high light conditions (Monotta and Pinzauti 1996).

Numerous studies of European beech (*Fagus sylvatica*) have demonstrated that seedlings respond to increased light with increased growth in height, diameter and overall biomass resulting in much larger individuals in gap locations and shifts in allometric relationships (e.g., Collet et al. 2002, Curt et al. 2005). However, European beech grows best at intermediate light levels of 10–25% full sunlight and in high light situations both high soil temperatures and soil drought can limit growth (Löf et al. 2005). In the long term, presence of overstory gaps was critical both for the maintenance of seedling banks (9–15% full sunlight requirement) and the ultimate release of growth restriction for seedling recruitment into the sapling size class (Szwagrzyk et al. 2001). European beech seedlings shift allocation patterns in response to available light and under higher light conditions more starch was stored in the root system as lateral root biomass (Gansert and Sprick 1998). Subsequently, Curt et al. (2005) found the increased allocation to the lateral roots of seedlings grown under higher light levels had a strong positive correlation with survival.

We expect that American beech seedlings may react similarly to above- and belowground manipulations as European beech. However, American beech appears morphologically and genetically isolated from other species in the genus (Denk et al. 2005), and this isolation may also be reflected in ecological differences at the seedling stage. We took

advantage of a mast event in autumn of 2002 to follow the effects of above- and belowground resources on growth and survival of naturally regenerating beech seedlings in HBEF in northern New Hampshire, USA. Natural canopy gaps and trenching were used to manipulate above- and belowground resources, respectively. Based on reviewed literature for both American and European beech seedlings, we propose the following hypotheses: 1) survivorship of a large cohort of beech seedlings following a mast year will be high, contributing to the seedling bank for this shade tolerant species, 2) gap openings will result in significantly larger American beech seedlings with higher survivorship and a shift in allometric relationships relative to seedlings grown under the closed canopy, and 3) any benefits of soil trenching to the seedlings will be most apparent in gap plots where light is not limiting and include increases in growth and tissue nitrogen concentrations but have little or no effect on survivorship. We also discuss our results in the context of long-term data for beech seedlings at HBEF, forest gap dynamics and BBD effects in northern hardwood forests.

Materials and Methods. **STUDY LOCATION AND DESIGN.** Hubbard Brook Experimental Forest (HBEF) is located in Woodstock, NH (43° 56' N, 71° 45' W) within the White Mountain National Forest. The HBEF has an area of 3160 ha and is covered mainly by northern hardwood forest dominated by American beech, sugar maple (*Acer saccharum* Marshall), and yellow birch (*Betula alleghaniensis* Britton) (Schwarz et al. 2003). In general, the regional climate is cool, temperate, humid continental. The mean July temperature is 19°C. Precipitation averages 1363 mm annually with around 40% of this precipitation in the form of snow. Snow pack typically persists at the HBEF from mid-November to mid-April (165 days, 30-yr average) with January air temperatures averaging –9°C. The soils are primarily well drained Spodosols with sandy loam to loamy sand textures. Soils are infertile and generally acidic (organic horizon pH ranges from 3.4–3.8). Additional details about this study site are available on the LTER website (<http://www.hubbardbrook.org/>).

Although the date of arrival of beech bark disease (BBD) in HBEF is not precisely

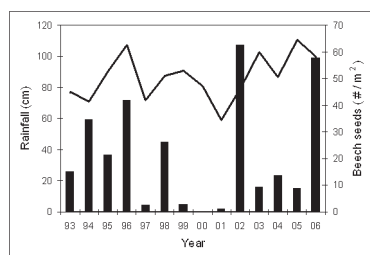


FIG. 1. Growing season rainfall and beech seed rain for the years 1993 to 2006 at Hubbard Brook Experimental Forest, NH. Beech seed rain is shown by the black bars with the scale given on the right y-axis. Rainfall data is depicted by the area line with the scale given in centimeters on the left y-axis. Beech seed data are from 51 long-term litter traps (each with a 0.096 m² catch area). Growing season rainfall was calculated from daily means of gauges in the south-facing watersheds and then summed for the snow-free period from mid-April to mid-November (C. Eagar and A. Bailey online LTER data).

known, BBD was not widespread in 1965 when vegetation studies commenced, however, BBD effects had become highly visible by 1977 (T.G. Siccama pers. comm.). Autumn 2002 was a beech “mast” year (year of unusually high seed production) in the northeastern US including HBEF (Fig. 1). Seed densities were calculated from 51 long-term litter traps (each with a 0.097 m² catch area) positioned at four elevations: 525, 585, 725, and 830 m in the Bear Brook watershed to the west of reference Watershed 6. The first years of growth for this cohort occurred against a backdrop of higher than average growing season rainfall (Fig. 1). Growing season rainfall was calculated from daily means of gauges in the south-facing watersheds and then summed for the snow-free period from mid-April to mid-November (C. Eagar and A. Bailey online LTER data: <http://www.hubbardbrook.org/data/dataset.php?id=13>).

The HBEF has a relatively low white-tailed deer density due in part to typically deep snow pack and hunting pressure (Pletscher et al. 1989). In the same study, beeches made up only 1.5% of the winter diets of white-tailed deer with no notable presence in their diets for the other three seasons, although the authors note that the study period (1976–1979) did not include any mast years. Herbivory pressure on seedlings is probably much lower in HBEF than in other northeastern sites with higher deer densities.

Three sites were chosen for this study in an area of the lower Hubbard Brook Valley with the highest beech basal area in the HBEF (Schwarz et al. 2003). The sites were selected for the presence of beech seedlings of the 2002 cohort. Sites ranged in elevation from 350 m (Site 1) and 400 m (Site 2) to 600 m (Site 3, just west of the base of W5). Slopes and aspects of the study areas within the sites are given in Table 1. Within each site, selection of study areas for the gap treatments was restricted to naturally formed canopy gaps created by the snap-off of one or two canopy dominant trees. These gaps were most likely cause by the 1998 ice storm (Rhoads et al. 2002) and were thought to be about five years old at the time of this study. Tree species for gap-makers were equally represented by beech, sugar maple and yellow birch with DBH mean (± 1 SD) values of 33.3 $\pm$ 8.9 cm. In order to ensure that the light treatment was effective, the nine natural gaps used in the study were amended by the removal of some sapling- and pole-sized trees within the actual gap. Gaps varied in shape, but the canopy openings (actual gap sensu Runkle 1982) ranged in area from 15–25 m² while the expanded gap (area at the ground-level that was affected by these openings was approximately 30–75 m²). Closed canopy study areas paired with the gap openings were selected to have similar seedling densities and to be in close proximity (5–10 m from the nearest expanded gap edge) and not north of the gap.

Within each canopy or gap study area, three 1 m² treatment plots were located based on spatial constraints (especially light exposure in the gaps) and presence of seedlings. In plots containing high sugar maple seedling densities, any sugar maple seedling touching a beech seedling was weeded out so as to limit interspecific competition within all plots. However, this was necessary in only a few of the plots. The trenching treatment was a trench roughly 20 cm deep around the plot perimeters. The trenches were initially dug in early June 2003, and were re-trenched in late July 2003, early September 2003, and July 2004.

VEGETATION AND ENVIRONMENTAL MEASUREMENTS. Vegetation and soil depth measurements were taken for each study area (level of canopy treatments) using nested circles centered in the middle of the study areas. Within a

Table 1. Summary of canopy treatment area characteristics measured at three study sites in the lower Hubbard Brook Valley, NH. For soil depth and PAR (photosynthetically active radiation), values are means and the numbers in parentheses denote 1 SD. For tree data, values in parentheses are the % of total value that is accounted for by beech. "Trees" are stems > 2 cm DBH. "Saplings" are stems > 1 m high, but < 2 cm DBH. The area names are given as site number, canopy treatment (S = canopy, G = gap), and area number. For example, 1G2 is replicate 2 of the gap amendment at Site 1.

Area name	Aspect (°)	Slope (%)	Soil depth (cm)	Rockiness (%)	Sunny day PAR (μmol sec ⁻¹ m ⁻²) mean (SD) max.		Sapling density (stems ha ⁻¹)	Tree density (stems ha ⁻¹)	Tree basal area (m ² ha ⁻¹)
1S1	190	32	28.9 (20)	39.02	12.6 (10.4)	43.9	955.4 (100)	780 (81.6)	10.6 (89.3)
1S2	220	9	22.0 (21)	41.46	9.5 (11.2)	48.7	331.2 (100)	716 (88.9)	23.2 (93.8)
1S3	258	17	30.1 (21)	39.02	14.6 (15.7)	69.2	509.6 (67.5)	637 (90.0)	10.8 (96.8)
1G1	192	31	18.0 (16)	2.44	30.4 (25.3)	92.2	229.3 (88.9)	732 (84.8)	21.5 (88.9)
1G2	220	18	37.8 (26)	43.90	35.6 (45.4)	175.5	382.2 (100)	796 (78.0)	11.4 (46.1)
1G3	258	20	28.5 (22)	17.07	204.1 (424.1)	1487	407.6 (84.4)	827 (88.5)	28.2 (97.7)
2S1	201	28	26.8 (19)	24.39	37.1 (52.9)	178.1	700.6 (100)	1321 (92.8)	28.6 (96.7)
2S2	184	21	32.1 (24)	29.27	24.8 (20.3)	69.2	700.6 (76.4)	748 (89.4)	16.4 (95.0)
2S3	150	16	39.5 (18)	2.43	23.6 (27.0)	108	267.5 (100)	875 (74.5)	19.4 (85.2)
2G1	171	24	32.9 (24)	41.46	119.9 (260)	1203	305.7 (70.8)	907 (64.9)	12.0 (74.0)
2G2	180	24	30.9 (18)	14.63	138.9 (210.8)	774.5	331.2 (96.2)	859 (72.2)	18.3 (93.0)
2G3	145	25	35.1 (21)	7.32	34.4 (36.4)	154.7	458.6 (97.2)	827 (78.8)	21.0 (84.8)
3S1	180	18	32.9 (18)	21.95	31.9 (41.5)	174.1	178.3 (85.7)	859 (74.1)	17.3 (88.5)
3S2	130	30	32.7 (20)	34.15	24.2 (25.6)	105.3	840.8 (93.9)	1003 (63.5)	20.5 (75.8)
3S3	160	19	29.3 (16)	41.46	23.7 (21.6)	84.0	458.6 (58.3)	875 (60.0)	24.0 (96.0)
3G1	158	14	30.4 (20)	2.44	98.6 (243.1)	1093	242.0 (36.8)	652 (56.1)	6.3 (75.4)
3G2	187	21	28.5 (20)	56.10	184.6 (347.9)	1520	535.0 (14.3)	621 (66.7)	8.1 (89.0)
3G3	163	10	18.1 (16)	58.54	128.6 (292.4)	1303	114.6 (66.7)	668 (57.1)	5.5 (70.4)

10 m radius circle, all trees greater than two centimeters in DBH were tallied by species and their DBHs were measured. Depth to obstruction (soil depth) was measured using a metal probe at every meter mark including the center point along the 10 m radius in each of the four cardinal directions (total of 41 measurements). Within a five meter radius circle, all woody stems taller than one meter, but less than two centimeters in DBH were tallied by species.

Light values for photosynthetically active radiation (PAR) were measured at the plot-level using an AccuPAR® Linear PAR/LAI ceptometer (Decagon Devices, Inc. Pullman, WA). Light measurements were taken at one meter height over the plots under cloudless conditions in mid-July 2003. In order to record the range of day light conditions, measurements were taken at four times of day: 0900, 1200, 1500, and 1700 hours. Also in mid-July 2003, soil moisture levels were assessed gravimetrically by taking three soil cores (two cm diameter) from within each plot. Samples were taken at least 48 hrs after the last rainfall event. Organic and mineral horizons were separated in the field and their lengths were recorded. The moist weight was taken and then the soil was dried for three days at 60°C before the dry weight was taken.

INITIAL ALLOMETRY MEASUREMENTS. In early June 2003, seedlings were destructively sampled outside of the study plots at all three sites to obtain measurements of initial seedling allometry. These seedlings were measured for above- and belowground biomass and length of the taproot.

In addition to the destructive samples, non-destructive measurements were taken on a subset of the seedlings in the study plots. The plots were subdivided into four 0.25 m² subplots and measurements were taken on all beech individuals from the NE subplot in early June 2003. Seedling height (from ground to bud tip), leaf length (following the mid-vein), and leaf width (widest point perpendicular to the mid-vein) were measured on these individuals in the field to the nearest millimeter with rulers. Leaf area was then estimated from length and width measurements using the equation:

$$\text{Leaf area} = \left(\frac{1}{2} \left(\pi * \left(\frac{1}{2} \text{ width} \right)^2 \right) + ((\text{length} - \frac{1}{2} \text{ width}) * \frac{1}{2} \text{ width}) \right)$$

This equation describes an area with a semicircular leaf base and triangular top of the leaf lamina. This shape roughly approximates the shape of a generalized beech leaf.

DENSITY COUNTS. Beech seedling densities were counted on 10 occasions over two growing seasons. In the first season (2003) counts were done bi-weekly from early June to early September (no mid-August count) and in the second season (2004) counts were taken monthly from early June to early September. Counts were made of the number of living and dead seedlings in the plots with a tally of healthy versus unhealthy living seedlings. An unhealthy seedling was defined as having > 10% of its leaf area missing or damaged. The apparent causes of damage in these seedlings were also recorded. The main sources of damage were: caterpillars, fungi, aphids, leaf miners, and mites. In the second season, we also tallied the number of seedlings whose buds had been eaten and those that failed to leaf out after winter.

MARKED INDIVIDUALS. Four individuals from each plot were chosen in a restricted random fashion for tagging in early June 2003 with the restrictions that: 1) each individual be located in a separate subplot and 2) two individuals be clumped and two solitary in order to examine any possible effects of intraspecific competition on the probability of seedling survival. Clumped seedlings had leaves that were touching the leaves of other beech seedlings, while the leaves of solitary seedlings did not touch those of any other seedlings. Measurements on the 144 marked individuals included stem heights, leaf lengths and widths, and leaf tracings that were taken once monthly in June, July, and August in both 2003 and 2004. Health and damage sources were also recorded as in the density counts.

Leaf tracings were scanned as black and white images at a resolution of 150 dpi. The images were then filled in using Adobe Photoshop version 6.0. Image areas were analyzed in UTHSCSA Image Tool for Windows version 3.0 and then converted from pixels to cm² by an empirically determined conversion factor.

SECOND SEASON HARVEST. In late September 2004, all seedlings in the control and trenched plots were harvested for analysis of above- and belowground seedling traits. Whole harvest of seedlings was attempted in all cases, however, some plants from each plot inevitably broke and these individuals were excluded from data

sets requiring both above- and belowground data. Seedlings were kept refrigerated until processing.

Aboveground traits recorded for each seedling included a damage code (0–5) for the amount of leaf area damaged. Damage categories were: 0 = top of seedling had been eaten, 1 = bud remained unbroken, 2 $\geq$ 75% of leaf area gone or damaged, 3 = 50–74% gone or damaged, 4 = 25–49%, and 5 < 25%. The numbers of leaves and buds were also recorded. Three stem lengths were taken: 1) radicle to the cotyledon scar, 2) 1st year growth, and 3) 2nd year growth to the bud tip. Forked stems were noted and their total lengths were summed. Leaves and stems were dried for 48 hrs at 60°C and weighed separately.

Lateral roots and tap roots were dried for 48 hrs at 60°C and weighed separately. Belowground traits also included the number of lateral roots and taproot length for the first 514 seedlings processed, but thereafter time constraints required processing for dry weights only. A subset of 24 seedlings (six per treatment combination) was ground for nitrogen concentrations. Percent of nitrogen and carbon in leaf tissues were determined by combustion on a Finnigan MAT Delta Plus continuous flow isotope ratio mass spectrophotometer (Cornell Stable Isotope Laboratory, Ithaca, NY).

FOUR-YEAR MARKED INDIVIDUALS. At the beginning of the experiment in each gap and canopy area, one of the three one m² plots was delineated, but otherwise unmanipulated. The purpose of these plots was to provide a control for manipulating seedlings (*sensu* Cahill et al. 2001). Therefore, density counts were made without touching and there were no marked seedlings or seedling measurements from these plots during the 2003 and 2004 growing seasons. After all seedlings were harvested from the manipulated plots in September 2004, the individuals remaining in the unmanipulated plots (total 316) were individually tagged and the height and number of leaves were recorded. For those surviving, height and number of buds were recorded in a final resurvey conducted October 2006.

STATISTICAL ANALYSES. In all analyses main effects variables regarded as fixed were the study area treatments (canopy and gap) and

plot-level treatments (control and trenched). Site and canopy treatment pairs were random effects. The design was fully nested with plots in canopy treatment within study areas within sites. Data sets with continuous response variables were analyzed using mixed linear models in SPSS 15.0. The terms: site, site by study area, and site by study area by canopy treatment were included as random variables to account for the nested design. For variables that were repeatedly taken during the study, date was included as a main fixed effect in the models and an extra random term (site by study area by canopy treatment by plot treatment) was included in the model.

The relationship of initial seedling density to site factors was examined using Proc Mixed in SAS 9.1 using the same model structure as described above for the mixed linear models, but including continuous site variables in the model statement. The best model was determined through an iterative process in which non-significant terms successively were removed from the model (beginning with the highest level interactions) and the model rerun.

The binary response of seedlings being either alive or dead at the end of the second and fourth field seasons was examined using Proc Genmod in SAS 9.1. Categorical variables for the two-year marked seedlings were: site, study area, canopy treatment (canopy or gap), plot treatment (control or trenched), seedling neighbors (clumped or solitary), and insect and fungal damage (present or absent). Seedling height and leaf area from the initial seedling measurements were included as a continuous explanatory variable in the model. Categorical variables for the four-year marked seedlings were: site, study area, and canopy treatment (canopy or gap). Continuous variables were seedling height and number of leaves in September 2004.

Differences in seedling allometry were explored using linear regression for two data sets: 1) one- and 16-month old beech seedlings from control closed canopy growing conditions and 2) 16-month old beech seedlings harvested in September 2004 that were grown under the canopy or in gap openings. For the first data set, only the relationship of below- to aboveground biomass was examined. For the second data set, six relationships were explored. All variable pairs were examined for the gain in the first variable with the increase

in the second variable: below- and above-ground mass, leaf mass and number, stem mass and height, taproot mass and depth, lateral and total root mass, and number of buds set and aboveground mass. Seedlings that had no leaves or were in the two greatest damage classes at the time of harvest were excluded from the data sets. For the second data set, seedlings were pooled for soil treatments since prior analyses had shown canopy treatment to be the main effect. Log₁₀ transformations of the variables did not significantly improve the fit of the regressions, therefore the results of the untransformed variable regressions are presented. Relationships were tested for significant differences in slopes of the lines using the 95% confidence intervals. Slopes with non-overlapping confidence intervals were considered significantly different at $\alpha = 0.05$.

Results. VEGETATION AND ENVIRONMENT. All study areas were dominated by beech, but also contained sugar maple, yellow birch, striped maple (*Acer pensylvanicum* L.), white ash (*Fraxinus americana* L.), and occasionally red spruce (*Picea rubens* Sarg.) and black cherry (*Prunus serotina* Ehrh.) with hobblebush (*Viburnum lantanoides* Michx.) in the understory. Study areas at Site 3 had a lower abundance of beech and greater prevalence of striped maple in the sapling layer (Table 1). Study areas were on south to southwest facing, moderate slopes (Table 1).

Light levels were significantly higher in gap areas than under the closed canopy ($F_{1,8} = 43.61$, $P < 0.001$). Soil depth to obstruction was fairly uniform across study areas ranging from about 20 to 40 cm, though Site 3 had the two rockiest study areas (Table 1). At the plot-level, soil moisture ($1.33 \pm 0.66\%$) and depth of the organic layer (4.72 ± 1.65 cm) did not differ significantly among treatments at the time of the July 2003 sampling (all P values > 0.05).

SEEDLING DENSITY AND SURVIVORSHIP. Within the study areas, initial seedling densities were best explained by the abundance of beech stems in the sapling layer ($F_{1,32} = 7.66$, $P = 0.009$) and percent slope ($F_{1,32} = 6.07$, $P = 0.019$). Densities were highest in flatter plots surrounded by the highest beech stem densities. At time zero (count in early June 2003 with already dead seedlings included), the 1 m²

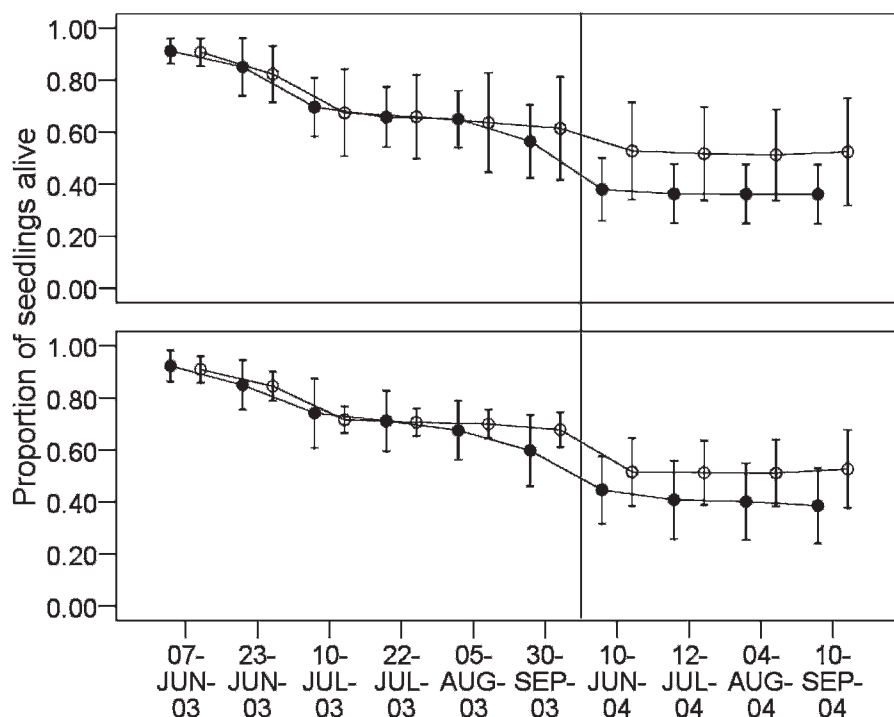


FIG. 2. Proportion of beech seedlings that survived for the first two growing seasons in plots where soil was not amended (top) or trenched (bottom) and the canopy was either closed (closed circles) or open by a small gap (open circles). The dashed line represents the break in surveys over the winter months. Error bars are ± 1 SD.

plots contained an average density of 52.18 ± 34.22 individuals (range: 8–183).

During the first growing season seedling survivorship reflected a significant interaction between canopy treatment and date with closed canopy plots having more deaths between the August and September sample times (Fig. 2, interaction: $F_{1,176} = 6.089$, $P = 0.015$). In general, date was the most significant predictor of surviving seedling proportions ($F_{1,176} = 493.0$, $P < 0.001$). In 2003, the largest declines in seedling numbers were recorded between early and late June (Fig. 2). In this growing season, leaf area tended to decrease over time (date: $F_{1,350} = 11.442$, $P < 0.001$) because of caterpillar herbivory and fungal browning. There was also a non-significant trend for greater leaf area loss on individuals in the trenched gap plots (Fig. 3).

Overwinter mortality the first winter was higher under the closed canopy (Fig. 2, $F_{1,28} = 7.24$, $P = 0.012$). The proportion of seedlings surviving overwinter was positively correlated (all $P < 0.05$, $N = 36$) to plot aspect ($r =$

0.345) and light ($r = 0.346$) and negatively correlated to plot rockiness ($r = -0.367$) and moisture of the organic layer in July 2003 ($r = -0.389$). During the second growing season, survivorship again varied mainly by time ($F_{1,104} = 58.757$, $P < 0.001$). Canopy treatment was close to significant (Fig. 2, $F_{1,28} = 3.502$, $P = 0.072$), as a result of the higher overwinter mortality in the canopy plots.

During 2004, the average number of leaves per seedling did not decrease significantly over time (Fig. 3), however, there was a significant site by canopy treatment interaction ($F_{2,6} = 51.4$, $P < 0.001$). Average leaf number was significantly higher in plots underneath gaps at all the sites, but the difference in the average leaf number between canopy treatments was greatest at Site 2 (Fig. 3). In general, leaf number appeared to be too coarse a scale for examining herbivory effects.

In both seasons, the two main sources of damage to seedlings were caterpillars and fungi. In 2003 damage tallies were taken on marked individuals ($N = 144$), and there were no significant differences in overall damage

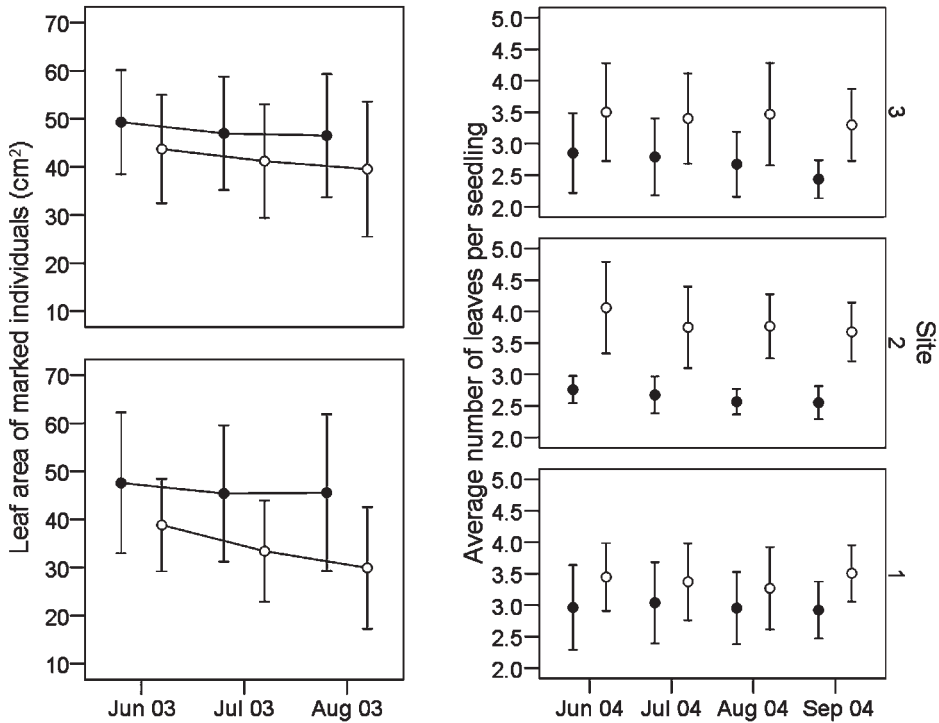


FIG. 3. Change in leaf area of marked individuals over time for the first growing season (left) and change in average leaf number per seedling for the second growing season (right). The canopy was either closed (closed circles) or open by a small gap (open circles). Error bars are ± 1 SD.

among treatments. Fungal damage was slightly higher ($57 \pm 30\%$) than caterpillar damage ($47 \pm 19\%$) in closed canopy plots while the reverse was true for gap plots (fungal: $46 \pm 25\%$; caterpillar: $61 \pm 16\%$). In 2004 damage to seedlings was assessed from plot survey data ($N = 889$ in June 2004; $N = 838$ by September 2004). Therefore, the 2004 data were not statistically comparable to 2003 data, however, fungal damage was clearly lower in 2004 ($24 \pm 17\%$) and caterpillar damage ($74 \pm 24\%$) was higher. The slight differences in prevalence of damage types between canopy and gap plots seen in 2003 were absent in 2004.

MARKED INDIVIDUALS. Two-year survival of marked seedlings was best explained by a model including site, canopy treatment and the interaction between initial seedling height and site (Fig. 4, Model fit: $X^2 = 21.147$, $df = 6$, $P = 0.002$, 70.1% predicted correctly). Survivorship was higher in plots at Site 1 ($df = 2$, Wald = 10.482, $P = 0.005$) and in gap plots ($df = 1$, Wald = 3.536, $P = 0.06$)

(Fig. 4). At Sites 1 and 3, taller seedlings were more likely to be dead by September 2004 (Fig. 4, $df = 2$, Wald = 9.210, $P = 0.01$).

In unmanipulated plots, the general patterns in seedling survivorship for the first two seasons were the same as shown for manipulated plots in Fig. 2 as there was no manipulation effect on two-year survival (comparison of manipulated and unmanipulated controls). At the end of the 1st season both canopy and gap unmanipulated plots had an average of 56% survival, and at the end of the second season canopy plots were at $40 \pm 23\%$ and gap plots were at $38 \pm 31\%$ survivorship. The percent surviving to the fourth year was $22 \pm 14\%$ in canopy plots and $30 \pm 27\%$ in gap plots.

Four-year survival of marked seedlings in the unmanipulated plots was best explained by a combination of seedling height, the interaction between seedling height and leaf number in September 2004, and the interaction between leaf number, canopy treatment and site (Fig. 5, Model fit: $X^2 = 60.221$, $df = 4$, $P < 0.001$, 71.7% predicted correctly). In contrast

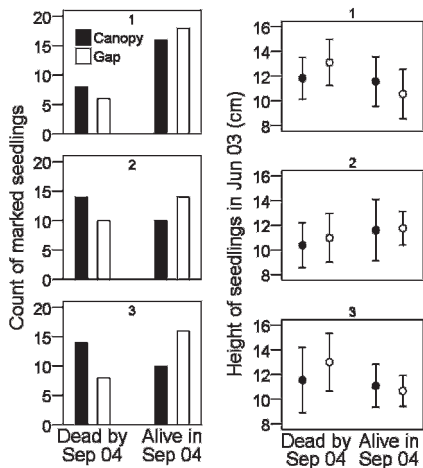


FIG. 4. Distribution of live and dead marked seedlings after two years as it relates to canopy treatment and seedling height in early June 2003. Data are paneled by Site numbers 1–3. Error bars are ± 1 SD.

to survivorship in the first two years, seedlings surviving from the second to the fourth year were significantly taller ($df = 1$, Wald = 5.368, $P = 0.021$). Individuals that died were more likely to have fewer leaves for their height

(Fig. 5, interaction: $df = 1$, Wald = 13.986, $P < 0.001$). The strength of the relationship between the number of leaves and living or dying was strongest for gap plots and plots at Site 3 (Fig. 5, interaction: $df = 2$, Wald = 7.433, $P = 0.024$). The pattern of larger seedlings in the gap plots continued into the 4th year and was consistent across sites (Table 2).

SEEDLING ALLOMETRY. Initial measurements taken on seedlings from each site outside of the plot areas show early season seedlings had aboveground biomass about four times greater than their belowground biomass (Table 3a). Taproots were significantly longer at Site 2 than at the other two sites ($F_{2,105} = 9.423$, $P < 0.001$, Tukey post hoc, $P < 0.05$). Leaf size measurements were all significantly larger for individuals in the closed canopy plots demonstrating the expected differentiation between sun and shade leaves (Table 3b). Initial within subplot seedling measurements show a canopy-gap by root trenching treatment interaction as trenched plots initially had taller seedlings in the canopy plots, but shorter seedlings in the gap areas (Table 3b), which was confirmed

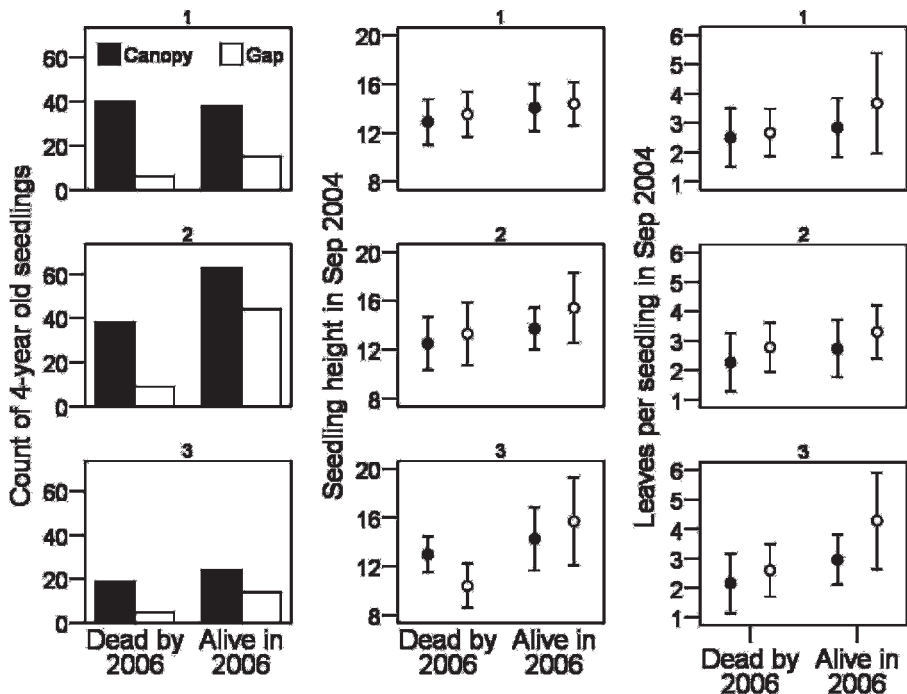


FIG. 5. Distribution of live and dead marked seedlings after four years as it relates to canopy treatment and seedling height and number of leaves per seedling in September 2004. Data are paneled by Site numbers 1–3. Error bars are ± 1 SD.

Table 2. Summary of measurements taken on 4-yr marked individuals in the unmanipulated plots at two and four years of age. All measurements were significantly different between canopy and gap plots ($P < 0.005$). Values are mean $\pm$ 1 SD.

Variable	Site 1		Site 2		Site 3	
	Canopy	Gap	Canopy	Gap	Canopy	Gap
Number of leaves						
Sept 04	2.67 $\pm$ 1.00	3.38 $\pm$ 1.56	2.55 $\pm$ 0.99	3.21 $\pm$ 0.91	2.60 $\pm$ 1.00	3.84 $\pm$ 1.64
Height Sept 04 (cm)	13.48 $\pm$ 1.99	14.13 $\pm$ 1.82	13.26 $\pm$ 2.00	15.06 $\pm$ 2.91	13.69 $\pm$ 2.24	14.30 $\pm$ 3.98
Height Oct 06 (cm)	16.94 $\pm$ 3.17	18.37 $\pm$ 3.59	16.77 $\pm$ 2.66	20.62 $\pm$ 6.51	17.52 $\pm$ 4.77	20.39 $\pm$ 4.27
2-yr height change (cm)	2.87 $\pm$ 2.30	3.99 $\pm$ 3.29	3.06 $\pm$ 1.88	5.19 $\pm$ 5.25	3.28 $\pm$ 3.27	4.70 $\pm$ 3.97
Number of buds						
Oct 06	1.29 $\pm$ 0.46	1.53 $\pm$ 0.83	1.24 $\pm$ 0.50	2.11 $\pm$ 1.48	1.50 $\pm$ 0.72	2.14 $\pm$ 1.29

by later measurements of 1st season stem lengths on harvested seedlings (Table 4).

By September 2004, the seedlings in the gap areas were larger than those from the closed canopy areas in almost all measurements both above- and belowground (Table 4). The number of buds was the only trait clearly related to both the canopy and soil treatments with more buds on seedlings from gap plots and trenched plots (Table 4). The number of lateral roots was lower in trenched plots under the canopy, but not in the gap plots.

Comparison of the relationship between above- and belowground biomass from one-month old seedlings (seedlings taken outside the plots in early June 2003) and 16-month old seedlings from canopy, control plots in late September 2004 showed a doubling of relative root biomass (Fig. 6). This greater increase in root biomass relative to aboveground biomass over time can be seen in the significant increase in the slope of this relationship (Fig. 6, slope (95%CI): for 1-month: 0.202 (0.157–0.247), 16-month: 0.373 (0.309–0.437), $P < 0.05$).

Gap opening significantly shifted the slope for the relationship for four of the six allometric relationships examined (Table 5). In three of these four cases, the slopes were steeper for seedlings in gap openings indicating greater gain in variable Y for each gain in X, which describes the larger size of gap-grown seedlings but also shows that gap seedlings had thicker (more weight per length) stems and roots than seedlings grown under closed canopy. For the fourth trait, seedlings grown under closed canopy had set more buds per mg of aboveground biomass than seedlings grown in gap openings by the end of the

second season (Table 5). The percent of seedlings with forked stems in their second season was observed to be about twice as high for gap-trenched plots (20.2%) as for the other plots (gap-control: 10.0%, canopy-control: 11.3%, and canopy-trenched: 8.4%).

Both trenching ($F_{1,12} = 13.98$, $P = 0.001$) and canopy treatment ($F_{1,12} = 6.58$, $P = 0.019$) had a significant effect on tissue N concentrations such that seedlings with highest tissue N were from trenched gap plots ($2.35 \pm 0.10\%$) and the lowest values were for those grown in closed canopy control plots ($1.97 \pm 0.18\%$) with the other treatments falling in between (canopy trenched: $2.08 \pm 0.19\%$, gap control: $2.05 \pm 0.05\%$).

Discussion. IMPORTANCE OF MAST SEEDING. As with other beech species, mast seeding appears to be important for maintaining a seedling bank in American beech. Overall survivorship of beech seedlings for our first season after the 2002 mast year was 59% (SE $\pm 2\%$), which compares closely to a study of *Fagus sylvatica* where two mast year cohorts had first season survival of 58% and 59% (Szwagrzyk et al. 2001). In their 10-year study, these authors were able to demonstrate that first season survivorship is higher in years with higher seedling densities. This agrees with the positive density dependence recently found for *F. grandifolia* seedling survival by Messaoud and Houle (2006) in Quebec. In general, first year survival of beech seedlings is generally lower in study areas prone to drought (Boerner and Brinkman 1996: Ohio 0%, Glitzenstein et al. 1986: Texas 7.3%) or deer herbivory (Liang and Seagle 2002: Maryland $> 30\%$).

Table 3. Summary of one-month old seedling traits (mean $\pm$ 1 SD) taken in early June 2003 at time of plot set-up in HBEF, northern NH.

a. Initial allometry measures from destructive samples taken on seedlings from outside of the plot areas. ** = significant difference between sites, $P < 0.001$.

Variable:	Site 1	Site 2	Site 3
Aboveground biomass (mg)	131.7 $\pm$ 33.9	141.2 $\pm$ 33.4	150.7 $\pm$ 40.1
Taproot length (cm)**	11.2 $\pm$ 3.8	14.8 $\pm$ 3.2	12.3 $\pm$ 3.8
Belowground biomass (mg)	34.8 $\pm$ 12.2	38.1 $\pm$ 9.1	38.7 $\pm$ 11.9

b. Measurements taken on all beech individuals from the NE 0.25 m² subplot in each plot. Leaf area was estimated from length and width measurements (see text for equation). Coding of significant effects: 1 = canopy-gap main effect, 3 = interaction of canopy-gap with soil treatment, all $p < 0.005$.

Variables	Canopy		Gap	
	Control	Trenched	Control	Trenched
Seedlings				
Height ³ (cm)	10.2 $\pm$ 2.4	10.9 $\pm$ 2.3	11.6 $\pm$ 2.3	10.0 $\pm$ 2.2
Leaf area ¹ (cm ²)	39.0 $\pm$ 12.5	39.3 $\pm$ 14.9	30.0 $\pm$ 14.5	31.9 $\pm$ 14.2
Leaf length ¹ (cm)	6.7 $\pm$ 1.3	6.8 $\pm$ 1.4	5.9 $\pm$ 1.6	6.2 $\pm$ 1.4
Leaf width ¹ (cm)	4.3 $\pm$ 0.9	4.4 $\pm$ 0.9	3.8 $\pm$ 1.1	4.0 $\pm$ 1.0

The beech seedling densities in plots at the beginning of this study (52.18 ± 34.22 stems per m²) were far greater than the majority of studies incorporating American beech seedlings (e.g., Ward 1961, Forcier 1973, Houle 1991, 1994, Thurston et al. 1992, Boerner and Brinkman 1996, Liang and Seagle 2002), however none of these studies followed mast events. One recent study in Mexico, similar to our study in following a mast event, found that beech seedling densities went from 8.8 stems per m² pre-mast to 51.5 stems per m² following a mast event in a forest dominated by *Fagus grandifolia* var. *mexicana* (Alvarez-Aquino and Williams-Linera 2002).

BEECH REGENERATION AT HBEF. Based on two previous studies that incorporated beech seedling counts, the experimental watersheds at the HBEF had a long-term average density of four beech stems of seed-origin per m² for the years 1965–1980 (Forcier 1973, Thurston et al. 1992) with a high of 10.7 ± 13.9 in 1970 on W4 in the year prior to clearcutting (Thurston et al. 1992). By the end of two years, densities in our manipulated plots were 20.9 ± 15 seedlings per m². In unmanipulated plots two year densities were comparable at 17.5 ± 17.5 seedlings per m² and by the end of four years densities were down to 11.0 ± 11.1 seedlings per m². The relatively high survivorship of seedlings at HBEF (Forcier 1973, this study) taken together with evidence for stable presence of beech stems of seed-origin in the

understory strongly suggest the presence of a beech seedling bank in this forest, although definitive proof would require a long-term demographic study (sensu Marks and Gardescu 1998).

In light of evidence presented in this study and previous studies, the beech seedling bank at HBEF may have contributed to patterns presently attributed to sprout-origin beeches in the literature. These patterns include the five-fold increase in saplings on W6 from 1965–1997 (T.G. Siccama unpublished data cited by Hane 2003), predicted responses of beech to the 1998 ice-storm (Rhoads et al. 2002), and patchiness of beech in the valley (Schwarz et al. 2003). The relative presence of seed-origin beeches may vary with topography as suggested by Bormann et al. (1970).

Seed-origin beeches may be more important at low elevations, while root sprouts may be more prevalent at higher elevations. Forcier (1973) did not find evidence for increased percentage of stems of sprout origin in the “seedling” size class with elevation and overall only 10.3 % of such stems were of sprout origin on Watershed 6 at the time of his study. However, BBD may have changed this pattern. Increased importance of sprouts at higher elevations may be probable through the combined effects of high winter mortality of seedlings and increased root sprouting in shallower soils where root injury tends to be more common. Winter mortality seems to be an important source of seedling mortality at

Table 4. Summary of measurements on 2-year old beech seedlings harvested in September 2004. Values shown are the mean $\pm$ 1 SD. Significant effects ($P < 0.05$) are coded with superscripts: 1 = canopy-gap main effect, 2 = soil treatment main effect, 3 = interaction of canopy-gap with soil treatment, ** = $P < 0.005$.

Variables:	Canopy		Gap	
	Control	Trenched	Control	Trenched
Counts (#):				
Leaves ^{1**}	2.14 $\pm$ 1.35	2.40 $\pm$ 1.31	3.16 $\pm$ 1.60	3.55 $\pm$ 1.85
Buds ^{1**, 2}	0.95 $\pm$ 0.75	1.15 $\pm$ 0.74	1.43 $\pm$ 0.74	1.74 $\pm$ 1.04
Lateral roots ³	60.39 $\pm$ 24.77	49.95 $\pm$ 20.54	59.40 $\pm$ 30.52	65.23 $\pm$ 29.81
Lengths (cm):				
Radicle-cotyledon ³	5.36 $\pm$ 1.62	5.73 $\pm$ 1.59	6.09 $\pm$ 1.65	5.60 $\pm$ 1.48
1 st -year stem ³	5.95 $\pm$ 1.34	6.10 $\pm$ 1.11	6.17 $\pm$ 1.17	5.80 $\pm$ 0.97
2 nd -year stem ^{1**}	3.19 $\pm$ 1.66	3.72 $\pm$ 2.35	4.68 $\pm$ 2.71	5.82 $\pm$ 3.66
Total stem ^{1**}	13.69 $\pm$ 3.35	15.00 $\pm$ 3.64	16.67 $\pm$ 3.24	16.86 $\pm$ 4.50
Longest root ¹	15.22 $\pm$ 4.05	16.30 $\pm$ 4.09	17.35 $\pm$ 5.16	18.29 $\pm$ 5.40
Weights (mg):				
Stems ^{1**}	120.3 $\pm$ 46.90	141.7 $\pm$ 64.18	204.2 $\pm$ 76.68	254.1 $\pm$ 164.9
Leaves ^{1**}	71.05 $\pm$ 73.92	77.39 $\pm$ 53.08	111.3 $\pm$ 58.94	128.9 $\pm$ 91.25
Total above ^{1**}	179.0 $\pm$ 104.3	210.2 $\pm$ 112.6	307.5 $\pm$ 134.0	375.0 $\pm$ 251.2
Tap root ^{1**}	66.62 $\pm$ 36.45	81.04 $\pm$ 55.93	148.1 $\pm$ 94.21	191.23 $\pm$ 146.7
Lateral roots ^{1**}	15.49 $\pm$ 12.52	16.75 $\pm$ 13.97	33.55 $\pm$ 26.91	44.80 $\pm$ 39.12
Total below ^{1**}	82.01 $\pm$ 44.62	97.26 $\pm$ 63.20	181.6 $\pm$ 111.6	236.0 $\pm$ 164.0
Unitless variables:				
Leaf damage ¹	3.50 $\pm$ 1.77	3.68 $\pm$ 1.59	4.26 $\pm$ 1.25	4.24 $\pm$ 1.11
Root:shoot ^{1**}	0.69 $\pm$ 0.28	0.69 $\pm$ 0.32	0.87 $\pm$ 0.41	0.96 $\pm$ 0.48

HBEF. Forcier (1973) found that 81% of the seedlings survived through the first growing season, but only 41% of the seedlings survived the first full year due to high over-winter mortality. Of those seedlings that died over-winter in this study, our observations indicate that $13 \pm 11\%$ died from small mammal herbivory of their buds and another $25 \pm 31\%$ failed to successfully leaf out in May 2004. At a nearby stand also in the White Mountains of New Hampshire, Graber and Leak (1992) noted failure of beech to reproduce from seed at 630 m elevation.

SEEDLING SIZE AND ALLOMETRY. Seedling height was an important predictor of seedling survivorship, which was also found in a recent study of European beech (Collet and Le Moguedec 2007). For the older European beeches of seed-origin, height, and diameter was positively related to survival. However in our study of very young seedlings, the relationship between seedling height and survival was complex and appeared to switch after the first two years. In the first two years, taller seedlings were more likely to die and this may partially explain the lack of a significant gap effect on survival during that time. After the second year, taller seedlings were more likely to

survive to be four-year old seedlings. We hypothesize that this switch may relate to the building up of secondary growth for structural support. That is, in earlier stages, taller seedlings would be more prone to bending and breaking, and many of these seedlings

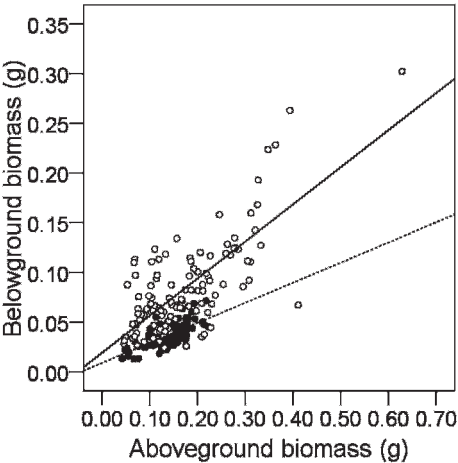


FIG. 6. Allometric relationships between above- and belowground biomass for beech seedlings at approximately one month old (early June 2003, black circles, lower line) and 16 months old (September 2004, white circles, upper line). Both regressions were highly significant ($P < 0.001$).

Table 5. Summary of regression analyses for allometric relationship differences between American beech seedlings grown under closed canopy and in gap openings. All linear regressions were highly significant with $P < 0.001$ and log-transformation of the variables did not significantly improve the fit of the relationships.

Variable		Closed canopy			Gap opening			Slope difference
X	Y	N	r ²	Slope (95% CI)	N	r ²	Slope (95% CI)	
Above wt.	Below wt.	281	0.656	0.47 (0.43–0.50)	248	0.572	0.56 (0.50–0.63)	*
Leaf wt.	Leaf no.	279	0.254	0.03 (0.02–0.03)	247	0.296	0.03 (0.02–0.03)	ns
Stem wt.	Stem height	281	0.579	14.62 (13.15–16.08)	248	0.628	25.77 (23.28–28.25)	**
Taproot wt.	Taproot length	181	0.196	4.38 (3.07–5.68)	233	0.270	12.19 (9.60–14.78)	**
Lateral root wt.	Below wt.	274	0.474	0.17 (0.15–0.19)	248	0.371	0.14 (0.12–0.19)	ns
Above wt.	Bud no.	282	0.333	3.71 (3.09–4.32)	249	0.292	2.46 (1.98–2.94)	*

were noted as bent or broken previous to or at their time of death. In contrast, once the seedlings have several years of woody growth, they would be less susceptible to any negative effects of being taller than their neighbors.

The shift in the relationship between above- and belowground biomass for seedlings between one- and 16-months of age reflects several important changes in the seedlings including greater structural complexity of the root system, senescence of cotyledons, and allocation of carbohydrates to roots in preparation for winter at the end of the second growing season. The greater biomass of both stems and taproots per unit length for gap-grown seedlings strongly suggests that at higher light levels the seedlings allocated more energy to secondary growth, although this hypothesis should be more accurately assessed using measurements of stem and root diameters. Collet et al. (2002) have shown this pattern for older European beech seedlings.

Latham (1992) observed that relative allocation by American beech seedlings to above- and belowground structures also shifts with light levels. At the lowest light levels beech allocated more to above- relative to belowground structures and this may predispose beech to drought sensitivity. Our 16-month old seedlings followed this pattern with lower root to shoot ratios under the closed canopy and a smaller gain in root biomass relative to gain in aboveground biomass. We did not find a relative increase in the contribution of lateral roots to total root biomass in gap-grown seedlings and this differs from the findings for European beech (Gansert and Sprick 1998, Curt et al. 2005). Another interesting allometric difference between seedlings grown under closed versus open canopies was the greater allocation of shade-grown seedlings to buds per unit of aboveground mass.

This pattern may correspond to the observation of Canham (1988) that beech saplings had no change in leaf area index between canopy and gap locations.

TRENCHING AND NITROGEN. In two year old seedlings, bud number was the only trait that appeared to be potentially co-limited by above- and belowground resources as both treatments had significant main effects for this plant trait. Although trenching had no statistically significant effect on seedling survival or size, it may be an important biological factor on infertile soils determining the ability of beeches to set bud and remain suppressed for long time periods in the understory. In our study, foliar nitrogen was highest for seedlings from trenched gap plots. Increased foliar nitrogen concentrations as a result of the root trenching treatments conducted on infertile soils appears to be a wide-spread phenomenon (Coomes and Grubb 2000, Ricard et al. 2003), however, the potential importance of nitrogen for bud set of tree species on infertile soils under low light conditions is less well documented.

This study agrees with several others on beech saplings that have shown primary light limitation for American beech growth and survival when the effects of canopy opening and soil N levels are compared (Finzi and Canham 2000, Ricard et al. 2003). However, soil N may play a more subtle role in determining the ability of beech to capitalize on increased light levels from gaps. This was suggested through effects on bud number and stem forking frequency in this study, and also by the increased number of leaves per shoot between beeches grown on fertile soils in Quebec (nine leaves average: Takahashi and Lechowicz 2008) and infertile

soils in New York (three leaves average: Canham 1988).

BEECH AS A GAP SPECIALIST. Based on studies of sapling recruitment to the canopy, American beech has been described as a small gap specialist (Runkle 1982, Canham 1988). Small gaps were also shown to positively affect growth rates of subcanopy beeches at HBEF (Poage and Peart 1993) and this study further supports the importance of small gaps for recruitment into the seedling bank. Seedlings in gap plots had higher survivorship and were larger both above- and belowground than under the closed forest canopy. The performance of seedlings in small gaps was highlighted by the finding of McClure et al. (2000) that individuals that established in gaps after gap formation were able to increase growth quickly and compete with initially larger individuals from advanced regeneration. We also observed this phenomenon on our plots as the gap-grown seedlings quickly overtopped older previously suppressed individuals (Cleavitt pers. obs.). For European beech, recruitment of seedlings to the sapling stage did not occur in the absence of gap light influence (Szwagrzyk et al. 2001). Canham (1990) found that suppression times for beech in the initial phase could range from no suppression up to 57 years before the first growth release. Our study suggests that individuals experiencing no initial suppression probably established in gaps.

In comparison with co-dominant sugar maple, beech was found to have shorter suppression times before recruitment to the canopy (Canham 1990). Beech has also been shown to benefit more from gradual canopy opening than sugar maple and this pattern has been attributed to the lower light threshold required by beech to experience growth release (Canham 1990, Nolet et al. 2008). Although sugar maple has been shown to increase radial growth in response to canopy opening from BBD effects (DiGregorio et al. 1999), beech may have a long-term advantage over sugar maple by achieving growth release in gradual gaps prior to actual gap formation (Krasny and Whitmore 1992, Krasny and DiGregorio 2001, Forrester et al. 2003). Recently, Takahashi and Lechowicz (2008) have emphasized that the relative importance of sugar maple and beech on the landscape probably relates to edaphic factors as well as gap dynamics.

BEECH REGENERATION POST-BBD. A key missing piece in beech demography at the HBEF and elsewhere is what percentage of the sapling sized beeches being seen in the wake of BBD were recruited from seed- versus sprout-origin. Hane (2004) has previously noted the lack of investigation into the origin of beech saplings seen to increase after BBD. However, the pattern used by her as proof of root-sprout origin (higher beech sapling densities surrounding heavily impacted beeches) could also have resulted at least in part from release of seed-origin beeches (i.e., seedling bank) in response to increased light levels under dying trees. Root sprouts are not necessarily more vigorous than seedlings. Indeed, both Forcier (1973) and Houston (2001) found root sprouts to experience mortality rates twice as high as those of seed-origin beeches in small size-classes. Both Canham (1984) and Takahashi and Lechowicz (2008) found negligible differences in recruitment patterns between saplings from the two origins. However, both studies also found that the majority of sapling sized beeches were of seed origin (Canham 1984 (New York), Takahashi and Lechowicz 2008 (Quebec)). In American beech, seedlings appear to have greater parasite immunity than root sprouts, which suffer from transmission of parasite susceptibility from their parent tree (Burt and Bell 1991).

The origin of the beech sapling increase noted in many studies (e.g., Houston 1975, Runkle 1990, 2007, Fahey 1998, Forrester et al. 2003, Hane 2003) is likely more complex than a mainly root sprout-origin at every site especially in light of: 1) recent population genetic studies on American beech that have shown comparatively high genetic diversity in "gray" and "red" beech populations known to vary in sprout production (Houston and Houston 2000, Kitamura and Kawano 2001, Morris et al. 2004), 2) realization that root sprouting is a response to root injury and may have been over-emphasized based on reference to forestry studies (e.g., Marquis 1964, Houston 1975), 3) support for American beech seedling banks in the literature (this study, Forcier 1973, 1975, Thurston et al. 1992, Houle 1991, 1994, Houston 2001), and 4) the ability of beech seedlings to benefit from gradual gaps and small canopy openings typical of BBD affected stands (this study, Poulson and Platt 1996, Beaudet and Messier 1998, Beaudet et al. 2000).

The relative contribution of seed- and root sprout-origin stems to the sapling size class for American beech within a site will depend on many factors including: 1) the relative importance of these two groups of progeny in the forest understory pre-BBD, 2) severity of BBD effects, 3) site specific factors that interact with BBD to promote root injury, 4) frequency of mast events, and 5) relative success of the two progeny types in recruiting to the sapling size class. Several recent studies have suggested that beech is becoming more dominant than sugar maple in the northern forests in response to canopy disturbances and that sprout regeneration by beech may be partially responsible for this pattern (Forrester et al. 2003, Beaudet et al. 2007). Future work will be needed to compare side-by-side the regeneration response of sprouts and seedlings to resource manipulations. This research need has also recently been emphasized by studies from northern beech populations in Quebec, which have developed hypotheses regarding the different roles of seedlings and root sprouts in beech regeneration (Beaudet et al. 1999, Arie and Lechowicz 2002). Such research would give a more complete picture of how these two regeneration strategies contribute to the relative importance of American beech in the northern hardwood forest under varying site conditions.

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